Number theoretic aspects of black holes in type IIB string theory on $K3 \times T^2$

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Outline

1. Black Holes Microstate Counting
2. Relation To Mock Modularity
3. Limitations
4. Generating Function of Single-Centered Degeneracy
5. Analytic Properties
6. Questions For Future

Black Hole Microstate Counting

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- One of the great successes of string theory: microscopic description of black holes. [Strominger-Vafa (1996), Dijkgraaf-Verlinde-Verlinde (1996), Shih-Strominger-Yin (2005), Gaiotto (2005), Shih-Yin (2005), Jatkar-Sen (2005), David-Jatkar-Sen (2006)].

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Find the Bekenstein-Hawking entropy of $\frac{1}{4}$-BPS black holes carrying electric and magnetic charges $(\vec{Q}, \vec{P})$ in Type IIB supergravity compactified on $\mathcal{M} \times S^1 \times \widetilde{S}^1$ with $\mathcal{M} = K3, T^4$.

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- Perfect match in large charge approx! [Reviews: Sen (2007), Mandal-Sen (2010)]

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- Degeneracy of $\frac{1}{4}$-BPS states with charges $(\vec{Q}, \vec{P})$ with $Q^2 = 2n, P^2 = 2m, Q \cdot P = \ell$ are coefficients of mock Jacobi forms of weight -10 and index m. [Dabholkar-Murthy-Zagier (DMZ) (2012)]

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- Generalization of both these connections to more general string compactifications called CHL models [Work-in-progress, RKS],[Bhand-RKS-Sen (2023), Banerjee-Bhutra-RKS (2025)]

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- For $N = 1, r = 22$. $N = 1, \mathcal{M} = K3$ is the DMZ case.

Degeneracy of $\frac{1}{4}$-BPS single centered black holes

Degeneracy formula

For charges Q, P with $Q^2 = 2m > 0, P^2 = 2n/N > 0, Q \cdot P = \ell$ with $4mn - N\ell^2 > 0$, the degeneracy is given by

$$d(m, n/N, \ell) = (-1)^{\ell+1} \int_{\mathcal{C}_{m,n,\ell}} d\tau d\sigma dz \, \frac{e^{-2\pi i(m\tau + n\sigma/N + \ell z)}}{\widetilde{\Phi}_k(\tau, z, \sigma)}$$

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where $\widetilde{\Phi}_k$ is a given meromorphic function, the contour $\mathcal{C}_{m,n,\ell}$ is given by

$$\mathcal{C}_{m,n,\ell} : \mathrm{Im}(\tau) = \frac{P^2}{N\varepsilon} \equiv \frac{2n}{N\varepsilon}, \quad \mathrm{Im}(\sigma) = \frac{NQ^2}{\varepsilon} \equiv \frac{2m}{\varepsilon},$$
$$\mathrm{Im}(z) = -\frac{P \cdot Q}{\varepsilon} \equiv -\frac{\ell}{\varepsilon}, \quad 0 \le \mathrm{Re}(\tau), \mathrm{Re}(z) < 1, \ 0 \le \mathrm{Re}(\sigma) < N \ ,$$

and $\varepsilon > 0$ is a small real number.

[Dijkgraaf-Verlinde-Verlinde (1994), David-Jatkar-Sen (2006), Cheng-Verlinde (2007)]

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- The choice $\mathcal{C}_{m,n,\ell}$ makes sure that only the single-centered black holes contribute to the degeneracy.
- At generic point in the moduli space, single and two-centered black holes exist, the choice $\mathcal{C}_{m,n,\ell}$ corresponds to a chamber of the moduli space where no two-centered black holes exist. More on this later.

Relation to Mock Modularity

Degeneracy formula revisited

- The function $\widetilde{\Phi}_k(\tau, z, \sigma)$ satisfies
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- Fourier-Jacobi expansion [Bhand-Sen-RKS (2023)]

$$\frac{1}{\widetilde{\Phi}_k(\tau, z, \sigma)} = \sum_{m=-\widetilde{\alpha}}^{\infty} \phi_m(\sigma, z) e^{2\pi i m \tau}, \quad \mathrm{Im}(\tau) \to \infty$$

$$\frac{1}{\widetilde{\Phi}_k(\tau, z, \sigma)} = \sum_{n=-\widetilde{\gamma} N}^{\infty} \psi_n(\tau, z) e^{2\pi i n \sigma/N}, \quad \mathrm{Im}(\sigma) \to \infty,$$

$\widetilde{\alpha}, \widetilde{\gamma} N \in \mathbb{Z}$.

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  $\widetilde{\alpha}, \widetilde{\gamma} N \in \mathbb{Z}$.
- Goal: express $d(m, n/N, \ell)$ in terms of integrals of ϕ_m, ψ_n. Need to deform the contour $\mathcal{C}_{m,n,\ell}$ to $\mathrm{Im}(\tau) \to \infty$ or $\mathrm{Im}(\sigma) \to \infty$.

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$\widetilde{\Phi}_k^{-1}$ has double poles on the hypersurface

$$n_2\left(\sigma\tau - z^2\right) + jz + n_1\sigma - m_1\tau + m_2 = 0$$

where

$$m_1 \in N\mathbb{Z}, \quad n_1 \in \mathbb{Z}, \quad j \in 2\mathbb{Z}+1, \quad m_2, n_2 \in \mathbb{Z}, \quad m_1n_1+m_2n_2+\frac{j^2}{4} = \frac{1}{4}.$$

The behaviour near $z = 0$ is

$$\widetilde{\Phi}_k(\tau, z, \sigma) = (e^{i\pi z} - e^{-i\pi z})^{-2} f^{(k)}(N\tau) g^{(k)}(\sigma/N) \ ,$$

for some modular forms $f^{(k)}, g^{(k)}$.

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DMZ case: $N = 1$, $\widetilde{\Phi}_k = \Phi_{10}$ (Igusa cusp form), $f^{(k)} = g^{(k)} = \eta^{-24}$.

Degeneracy formula revisited

- There is a special set of charges A, B such that the contour $\mathcal{C}_{m,n,\ell}$ can be deformed to $\mathrm{Im}(\tau) \to \infty$, $\mathrm{Im}(\sigma) \to \infty$ respectively without picking up poles [Bhand-Sen-RKS (2023)]

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- Doing some straightforward calculation after deforming the contour [Dabholkar-Murty-Zagier (2012),Bhand-Sen-RKS (2023)]

$$d\,(m, n/N, \ell) = (-1)^{\ell+1} \int d\sigma \int dz e^{-2\pi i \frac{n}{N}\sigma} e^{-2\pi i \ell z} \phi^F_m(\sigma, z)\ , \quad (m, n, \ell) \in A$$

where

$$\phi^F_m(\sigma, z) = \sum_{\ell \in \mathbb{Z}} e^{2\pi i \ell^2 \sigma/4m} e^{2\pi i \ell z} h_\ell^{\left(-\frac{\ell}{2m}\sigma\right)}(\sigma)\ ,$$

and

$$h_\ell^{\left(-\frac{\ell}{2m}\sigma\right)}(\sigma) := e^{-2\pi i \ell^2 \sigma/4m} \int_{-\frac{\ell}{2m}\sigma}^{-\frac{\ell}{2m}\sigma+1} dz e^{-2\pi i \ell z} \phi_m(\sigma, z)\ .$$

Degeneracy formula revisited

For $(m,n,\ell) \in B$

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Conclusion

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Generalization to $\mathbb{Z}_M$-invariant states in $\mathbb{Z}_N$ CHL models [BSB (2025)].

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Using this, one can prove that

$$\phi_m\left(\frac{a\sigma+b}{c\sigma+d}, \frac{z}{c\sigma+d}\right) = (c\sigma+d)^{-k} e^{2\pi i\left(\frac{mcz^2}{c\sigma+d}\right)} \phi_m(\sigma,z) \ , \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma^1(N)$$

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These are meromorphic Jacobi forms of weight $-k$ and index $m, n/N$ respectively.

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Compare with the simplest example:

$$E_2(\tau) = 1 - 24 \sum_{n=1}^{\infty} \sigma_1(n) e^{2\pi i n \tau} .$$

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But if we define $\widehat{E}_2(\tau) = E_2(\tau) - \frac{3}{\pi \mathrm{Im}(\tau)}$, then

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- S-duality invariance of the degeneracy is obscure.
- Goal: construct the generating function of single-centered degeneracy which is manifestly duality invariant.

Generating Function of Single-Centered Degeneracy

The generating function

We want to construct the function

$$F(\tau,\sigma,z)=\sum_{\substack{m,n,\ell\in\mathbb{Z}\\ m,n\geq 0,4mn-N\ell^2\geq 0}} d(m,n/N,\ell)e^{2\pi i(m\tau+n\sigma/N+\ell z)}\ ,$$

where $\tau,\sigma\in\mathbb{H}$, $z\in\mathbb{C}$, $\tau_2\sigma_2-z_2^2>0$ (Siegel upper half space $\mathbb{H}_2$)

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- When $4mn-N\ell^2=0$, deform the contour as follows [Bhand-Sen-RKS (2025)]

$$\mathrm{Im}(\tau) = \frac{2n+\varepsilon_1}{N\varepsilon} \ , \quad \mathrm{Im}(\sigma) = \frac{2m+\varepsilon_2}{\varepsilon} \ , \quad \mathrm{Im}(z) = -\frac{\ell+\varepsilon_3}{\varepsilon} \ ,$$

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- Gives unambiguous answer independent of the way we deform the contour.
- Goal: show that this generating function converges in some domain of $\mathbb{H}_2$.

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- Relate $F(\tau, \sigma, z)$ to $\widetilde{\Phi}_k$ and use analytic continuation.

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- Focus on DMZ $N = 1$ case.

Two-centered configurations and wall crossing

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- In terms of BPS states, two-centered black holes correspond to those $\frac{1}{4}$-BPS states which are unstable against decay into half-BPS states.
- All two-centered configurations have charges of the form

$$Q = aM + cN \ , \quad P = bM + dN \ , \quad ad - bc = 1 \ ,$$

corresponding to the bound state of half BPS states with charges (aM, bM) and (cN, dN) which are S-duality equivalent to small black holes with charges $(M, 0), (0, N)$ [Dabholkar-Gaiotto-Nampuri (2007), Sen (2007)].

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- On the other hand, the poles of Φ_{10}^{-1} with $n_2 = 0$ divides $\mathbb{H}_2$ into chambers:

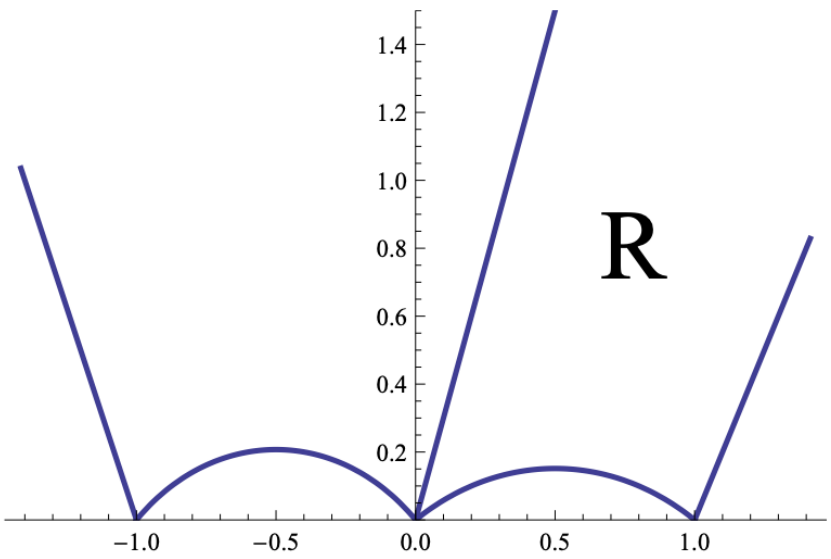
$$-m_1 + n_1 (\sigma_2/\tau_2) + j (z_2/\tau_2) = 0, \quad m_1, n_1, j \in \mathbb{Z}, \quad m_1 n_1 + j^2/4 = 1/4$$

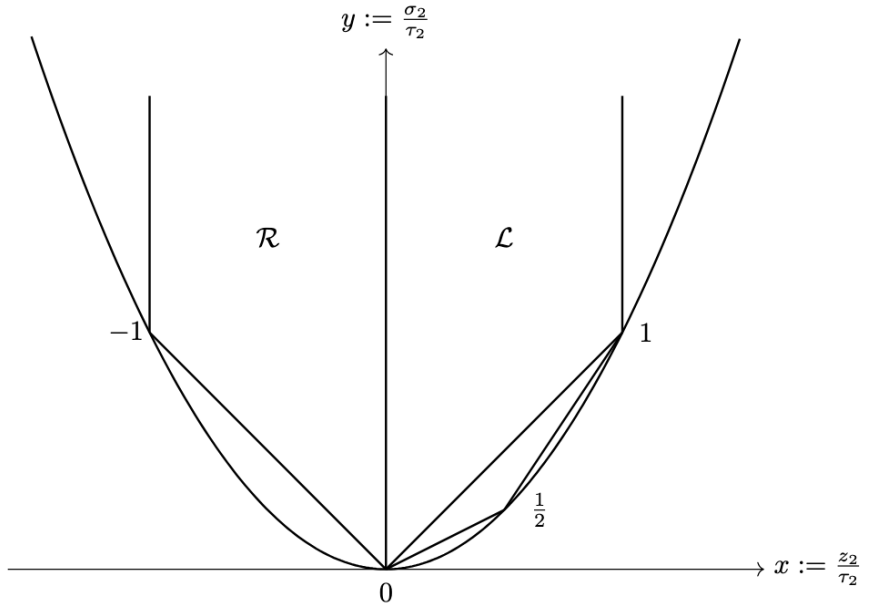
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- There is a one-one correspondence between chambers in τ plane and chambers in $\mathbb{H}_2$ [Sen (2011)]





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- For $P^2 = Q^2 = -2$, infinitely many identifications.
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- Justification: Supergravity analysis [Chowdhuri-Lal-Saha-Sen (2012)].

Degeneracy of two-centered configuration

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- The generating function in a given chamber is then given by

$$(e^{\pi i z} - e^{-\pi i z})^{-2} f(\tau) f(\sigma) \ .$$

Naive guess for generating function

To get only single-centered contributions, naively one would subtract $(e^{\pi i z} - e^{-\pi i z})^{-2} f(\tau) f(\sigma)$ and its $\mathrm{SL}(2,\mathbb{Z})$-images from $1/\Phi_{10}$:

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The subseries corresponding to $(a,b,c,d) = (1,0,c,1), c \in \mathbb{Z}$ has a divergent contribution:

$$\sum_{c \in \mathbb{Z}} (e^{\pi i (c\tau + z)} - e^{-\pi i (c\tau + z)})^{-2} e^{-2\pi i (c^2\tau + \sigma + 2cz)} f(\tau) \ .$$

Recognizing the two-centered contributions

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Using the expansion

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we can write the subtraction term as

$$-\frac{1}{2} \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{PSL}(2,\mathbb{Z})} \sum_{r>0} r \Big\{ e^{\pi i r(ac\tau + bd\sigma + (ad+bc)z)} H(ac\tau_2 + bd\sigma_2 + (ad+bc)z_2)$$

$$e^{-\pi i r(ac\tau + bd\sigma + (ad+bc)z)} H(-ac\tau_2 - bd\sigma_2 - (ad+bc)z_2) \Big\}$$

$$\sum_{p,q \geq -1} f_p f_q e^{2\pi i p(a^2\tau + b^2\sigma + 2abz)} e^{2\pi i q(c^2\tau + d^2\sigma + 2cdz)}$$

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Focus on $p, q \geq 0$ terms.

Recognizing the two-centered contributions: $p, q \geq 0$

The $p, q \geq 0$ terms can be reorganized as

$$-\frac{1}{2}\sum_{\left(\begin{smallmatrix}a & b\\ c & d\end{smallmatrix}\right)\in \mathrm{PSL}(2,\mathbb{Z})}\sum_{r>0}\sum_{p,q\geq 0} rH\left(ac\tau_2+bd\sigma_2+(ad+bc)z_2\right) f_p f_q$$

$$\times e^{2\pi i\left\{\left(pa^2+qc^2+rac\right)\tau+\left(pb^2+qd^2+rbd\right)\sigma+(2pab+2qcd+r(ad+bc))z\right\}}$$

$$-\frac{1}{2}\sum_{\left(\begin{smallmatrix}a & b\\ c & d\end{smallmatrix}\right)\in \mathrm{PSL}(2,\mathbb{Z})}\sum_{r>0}\sum_{p,q\geq 0} rH\left(-ac\tau_2-bd\sigma_2-(ad+bc)z_2\right) f_p f_q$$

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Recognizing the two-centered contributions: $p, q \geq 0$

The terms containing red exponents can be recognized as the contribution coming from the bound state of half-BPS states carrying charges (aM, bM) and (cN, dN) satisfying

$$M^2 = 2p \ , \quad N^2 = 2q \ , \quad M \cdot N = r, \quad r > 0 \ ,$$

so that the total charge $(Q, P) = (aM + cN, bM + dN)$ satisfies

$$Q^2 = 2(pa^2 + qc^2 + rac) \ , \quad P^2 = 2(pb^2 + qd^2 + rbd) \ ,$$
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Such bound states exist in the chamber $ac\tau_2 + bd\sigma_2 + (ad + bc)z_2 > 0$ for $r > 0$ [Sen (2011)].

Recognizing the two-centered contributions: $p, q \geq 0$

The terms containing blue exponents can be recognized as the contribution coming from the bound state of half-BPS states carrying charges (aM, bM) and (cN, dN) satisfying

$$M^2 = 2p \ , \quad N^2 = 2q \ , \quad M \cdot N = -r, \quad r > 0 \ ,$$

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Such bound states exist in the chamber $ac\tau_2 + bd\sigma_2 + (ad + bc)z_2 < 0$ for $r > 0$ [Sen (2011)].
Thus these subtractions are correct.

Recognizing the two-centered contributions: $p = -1, q \geq 0$ and $p \geq 0, q = -1$

Reorganize the sum to write

$$
\begin{aligned}
&- \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{PSL}(2,\mathbb{Z})} \sum_{r>0} \sum_{p \geq 0} f_p f_{-1} r H\left(ac\tau_2 + bd\sigma_2 + (ad+bc)z_2\right) \\
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\end{aligned}
$$

Used $(a, b, c, d) \to (-c, -d, a, b)$ and $p \leftrightarrow q$ symmetry to drop the $1/2$ and $f_{-1} f_q$ terms.

Recognizing the two-centered contributions: $p = -1, q \geq 0$ and $p \geq 0, q = -1$

These terms represent subtraction of the contribution from the bound states of half-BPS states carrying charges $(aM, bM), (cN, dN)$ satisfying

$$M^2 = 2p \ , \quad N^2 = -2 \ , \quad M \cdot N = r, -r, \quad r > 0 \ ,$$

so that the total charge $(Q, P) = (aM + cN, bM + dN)$ satisfies

$$\begin{aligned} Q^2 = 2(pa^2 - c^2 + rac) \ , \quad P^2 = 2(pb^2 - d^2 + rbd) \ , \\ Q \cdot P = 2pab - 2bd + r(ad + bc) \ , \\ Q^2 = 2(pa^2 - c^2 - rac) \ , \quad P^2 = 2(pb^2 - d^2 - rbd) \ , \\ Q \cdot P = 2pab - 2bd - r(ad + bc) \ . \end{aligned}$$

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- BSM tells us to identify these two contributions and it is nontrivial only if it exists in both the chambers corresponding to (a, b, c, d) and $(a, b, c - ar, d - br)$.
- Thus we must replace the $p = -1, q \geq 0, p \geq 0, q = -1$ sum by

$$
\begin{aligned}
&- \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{PSL}(2,\mathbb{Z})} \sum_{r>0} r \sum_{p \geq 0} f_p f_{-1} H(ac\tau_2 + bd\sigma_2 + (ad + bc)z_2) \\
&\times \; H\left(-ac\tau_2 - bd\sigma_2 - (ad + bc)z_2 + ra^2\tau_2 + rb^2\sigma_2 + 2rabz_2\right) \\
&e^{2\pi i\{(pa^2 - c^2 + rac)\tau + (pb^2 - d^2 + rbd)\sigma + (2pab - 2cd + r(ad + bc))z\}}
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The subtraction term can be written as

$$-f_{-1}^2 \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{PSL}(2,\mathbb{Z})} \sum_{r>0} r H\left(ac\tau_2 + bd\sigma_2 + (ad+bc)z_2\right)$$
$$\times\, e^{2\pi i\left\{\left(-a^2-c^2+rac\right)\tau+\left(-b^2-d^2+rbd\right)\sigma+(-2ab-2cd+r(ad+bc))z\right\}}.$$

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These terms represent subtraction of the contribution from the bound states of half-BPS states carrying charges $(aM, bM), (cN, dN)$ satisfying

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$$Q^2 = 2(-a^2 - c^2 + rac)\ , \quad P^2 = 2(-b^2 - d^2 + rbd)\ ,$$
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- The exponent in the $p = q = -1$ sum remains unchanged under $(a, b, c, d) \to (a_n, b_n, c_n, d_n)$, where

$$\begin{pmatrix} a_n & b_n \\ c_n & d_n \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & -r \end{pmatrix}^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} \ , \quad n \in \mathbb{Z} \ .$$

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Thus we must replace the sum by

$$-f_{-1}^2 \sum_{r>0} r \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in G_r \backslash \mathrm{PSL}(2,\mathbb{Z})} \prod_{n=-\infty}^{\infty} H(a_n c_n \tau_2 + b_n d_n \sigma_2 + (a_n d_n + b_n c_n) z_2)$$
$$\times \; e^{2\pi i \{(-a^2 - c^2 + rac)\tau + (-b^2 - d^2 + rbd)\sigma + (-2ab - 2cd + r(ad+bc))z\}} ,$$

where G_r is the cyclic group generated by $\left(\begin{smallmatrix} 0 & -1 \\ 1 & -r \end{smallmatrix}\right)$.

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$\widetilde{F}(\tau,\sigma,z) = \frac{1}{\Phi_{10}} - S(\tau,\sigma,z)$ where [Bhand-Sen-RKS (2025)]

$$S(\tau,\sigma,z) = \frac{1}{2} \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{PSL}(2,\mathbb{Z})} f_+(a^2\tau + b^2\sigma + 2abz)\ f_+(c^2\tau + d^2\sigma + 2cdz)$$

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Analytic Properties

Properties of $\widetilde{F}$

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Theorem (Convergence of the sum) [Bhand-Sen-RKS (2025)]

The sum over a, b, c, d, r in S converges absolutely and uniformly on compact subsets of $\mathbb{H}_2$ except on the hypersurface

$$m_1\tau - n_1\sigma + m_2 + j\,z = 0, \quad m_1, n_1, m_2 \in \mathbb{Z}, \quad m_1 n_1 + \frac{j^2}{4} = \frac{1}{4}\,,$$

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Holomorphicity of $\widetilde{F}$ [Bhand-Sen-RKS (2025)]

$\widetilde{F}$ is holomorphic in the domain $\det \mathrm{Im}\,\Omega > \frac{1}{4}$, $\Omega = \begin{pmatrix} \tau & z \\ z & \sigma \end{pmatrix}$ and has double poles on the hypersurface

$$n_2(\tau\sigma - z^2) + m_1\tau - n_1\sigma + m_2 + j\,z = 0,$$

$$m_1, n_1, m_2, n_2 \in \mathbb{Z}, \qquad n_2 \geq 1, \qquad m_1 n_1 + m_2 n_2 + \frac{j^2}{4} = \frac{1}{4}\,.$$

Main Theorem

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Analytic continuation of F [Bhand-Sen-RKS (2025)]

We have $F = \widetilde{F}$ and hence the generating function $F(\tau, \sigma, z)$ admits a meromorphic continuation to $\mathbb{H}_2$.

The generating function is thus manifestly S-duality invariant:

$$F(\gamma\Omega\gamma^t) = F(\Omega) \ , \quad \Omega = \begin{pmatrix} \tau & z \\ z & \sigma \end{pmatrix} \in \mathbb{H}_2 \ , \quad \gamma \in \mathrm{SL}(2,\mathbb{Z}) \ .$$

Questions For Future

- Does there exist a nonholomorphic function F^* such that $F + F^*$ transforms as a Siegel modular form for $\mathrm{Sp}(2, \mathbb{Z})$?

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- If these two statements are true, then following the standard terminology of mock modular forms, we may call F a mock Siegel modular form.
- Generalize the story to CHL models $N > 1$.

Thank You